

Part 1: Label and Analyze the Data

Part 2: Train a Model

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Reading: Your Tasks for Part 2
5 min
- 📄

Graded Assignment: Training a Model
1h
- 🔒

Reading: Our trained model
5 min

Part 3: Evaluate Your Model

Summary and Next Steps

Your Tasks for Part 2

Use the code examples you have from earlier in this course to train a detection model for the parking signs dataset. Because training detection models usually requires significant resources and time, we provide some recommended training options to help you. Using these settings, you should be able to train a model in about 30 minutes on a CPU.

If you have some patience and a computer with more memory (or a supported GPU), we recommend training additional models with more optimal options to get better results.

***Do not* resize the images for training.** Load one of the images into MATLAB to determine the size and use that as your training size.

Recommended Options (see below for an explanation for each recommendation)

1. Use YoloX with the tiny-coco backbone (no anchor boxes required)
2. Use a mini-batch size of 16
3. Train for 10 epochs
4. Set a validation frequency so the validation loss is calculated at least once every epoch. This will depend on how many images you use for training and your mini-batch size.

You may find the [Common Training Options Reference](#)  page from course one helpful in selecting training options.

Use YOLOX with the tiny-coco backbone

YOLOX represents the most up-to-date and memory-efficient object detection model in MATLAB (release R2023b). Using the tiny-coco backbone also reduces the training time and inference time. Another benefit of YOLOX is that it does not require anchor boxes, simplifying your model setup.

If you're curious, try training a YOLOv4 model and compare the results. YOLO v4 with the tiny-yolo backbone also trains in a reasonable amount of time.

Use a mini-batch size of 16

Available memory is often a limiting factor for mini-batch size when training detection models. We achieved good results using a mini-batch size of just 16, which reduces the memory requirements significantly. This approach works here because there are only 400 training images and just 3 classes.

You are encouraged to try doubling, or even tripling the mini-batch size if your computer has more than 16 GB of RAM.

Train for 10 epochs

Training for 10 epochs was enough for us to achieve good results. Our loss curve was still decreasing at 10 epochs, but it was decreasing slowly. If you are willing to wait longer or have access to a GPU, we encourage you to increase this value.

Go to next item ✓ Completed

